

Adam Jemmali

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Professional Summary

Backend focused computer science student with hands on experience building production-ready Python/Fast API services, data pipelines and AI workflows with Make.com and n8n seeking a Backend Engineer Intern role. Strong foundation in backend systems, APIs, databases and cloud deployment, with applied experience integrating machine learning into real-world products.

Education

University of Ottawa <i>Bachedatabases andScience</i>	Ottawa, ON Sep 2024 – Present
Dawson College <i>DEC in Pure & Applied Science</i>	Montreal, QC Jan 2023 – May 2024

Experience

Freelance AI Automation Specialist Upwork	May 2024 – Present Remote
- Designed and deployed Python-based backend services using FastAPI to power automation workflows for real estate and recruiting teams, improving qualified lead conversion by 35%. - Built API-first data pipelines integrating CRMs, Slack, and third-party services with retries, validation, and failure handling, achieving 40% ROI within 3 months. - Integrated lightweight ML and LLM-based classification modules into live workflows while maintaining low latency and system reliability.	
Software Developer Hoopsalytics	May 2024 – Aug 2024 Montreal, QC
- Designed and implemented a FastAPI backend for video ingestion, processing and inference workflows, reducing runtime errors by 25%. - Built and optimized a computer vision inference pipeline using OpenCV and MediaPipe to detect basketball events with 92% accuracy. - Exposed backend analytics via REST APIs and collaborated on a React dashboard, improving internal analysis workflows and reducing manual review time by 30%.	
Office IT Support Specialist Ad. Systemes S.A.R.L	Jul 2018 – Aug 2018 Casablanca, Morocco
Provided hands-on technical support across hardware, OS, and networking issues, improving system uptime and operational consistency.	

Projects

Chronos Cloud <i>Python, NATS, MongoDB, Next.js, Gemini, OC Transpo, Solana, Eleven Labs</i>	Jan 2026
Architected an event-driven digital twin using NATS with JetStream, implementing durable messaging and guaranteed retries to simulate real-time crisis response across complex transit and power domains.	

- **Engineered a FastAPI control plane** to host modular agentic frameworks, utilizing **circuit-breaker patterns to ensure deterministic fallbacks during LLM latency, prioritizing system stability over AI availability.**
- **Integrated live public data APIs** (OC Transpo, transit feeds) into a fault-tolerant pipeline, achieving 92% simulation accuracy while managing high-concurrency event streams.

OmniContext OS | *Backboard.io, Make.com, Slack, Vector Stores, LLMs*

Jan 2026

- **Architected a provider-agnostic memory layer** using **FastAPI**, abstracting vector storage (ChromaDB/Pinecone) behind a clean interface to ensure infrastructure flexibility.
- **Engineered async ingestion pipelines** with **FastAPI's native asyncio support**, handling high-throughput Slack and Email data with custom backpressure and rate-limiting to maintain predictable retrieval latency.
- **Designed a multi-tenant metadata schema** for scoped retrieval, enabling time-aware context assembly and workspace isolation across fragmented communication channels validated by a **top 9 finish** hackathon teams.

Online Tutoring App Management System (MVP) | *Java, XML, Android SDK, SQLite, Firebase, UML*

Sep 2025-Dec 2025

- Led end-to-end UML system design for a multi-role platform (Student, Tutor, Admin) with strict access boundaries.
- Implemented layered Android architecture with Singleton pattern and client-role pattern management, delivering booking, approvals, cancellations, and rating features.
- Delivered iterative features using Agile sprints and GitHub releases, reducing development iteration time by 30%.
- **Orchestrated a PostgreSQL (Supabase) backend** with strict multi-role access boundaries, managing complex relational data for real-time booking and approval features.
- **Implemented JUnit testing suites** for input validation and business logic, ensuring system stability and reducing development iteration time by 30%.

Skills

Backend: Python, Go, Java, C++ FastAPI, REST APIs, Async Programming, PostgreSQL, Redis

Computer Science fundamentals: Data Structures, Algorithms, Big-O Analysis, Discrete Maths, Linear Algebra

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Supabase, Firebase

DATA & AI & ML: pandas, NumPy, OpenCV, MediaPipe

Infrastructure & DevOps: Docker, Linux, AWS, Git, GitHub Actions, Kubernetes

Testing & Tooling: Pytest, JUnit, Postman, ThunderClient

AI Automation: Make.com, Zapier, N8N